



JOHNS HOPKINS
UNIVERSITY

ARCH Portal

User Guide

A role-based reference

User - PI - Staff - Superuser - Admin

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Advanced Research Computing at Hopkins (ARCH)



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Find your role

ARCH Portal (CLOACK Stack) serves five distinct audiences. Each role guide is self-contained for that audience and links into the deep references when you need more.

Who are you?

If you are...	...read	...in one line
A researcher submitting jobs (JHED, `ext-*`, or `ssci-*`)	[`user.md`](user.md)	How to log in, submit `sbatch`, check `sbalance`, read your usage
A PI managing a project, its allocations, and billing	[`pi.md`](pi.md)	How to request a paid allocation, add users, set weekly caps, read the billing report
Helpdesk staff handling tickets	[`staff.md`](staff.md)	How to triage, route by skill, escalate, hit FQR / SLA, run analytics
A ColdFront / Django superuser (cluster ops + identity)	[`superuser.md`](superuser.md)	How to approve allocations, drain nodes, reset OTP, rotate the `/admin/` password
A cluster / DevOps admin running `start.sh` and Ansible	[`admin.md`](admin.md)	How to build, publish, promote, and deploy the 4-stage pipeline

Not sure which one applies? The roles are not mutually exclusive a PI may also be a user; a superuser may also be an admin. Read the role(s) that match what you are doing right now, then follow the cross-links.

What lives where

- In-app help [/user/help/](#) (all signed-in users) and [/staff/help/](#) (Helpdesk staff / superusers) are the live, admin-editable surface for short-form operational notes. They overlap with this directory by design: these role guides are the offline / pre-login / printable reference, the in-app pages are the live working surface.
- Deep technical docs [slurm/README.md](#), [coldfront/BILLING_SETUP_GUIDE.md](#), the top-level **README.md**, and **CLOACK.md** cover architecture and the full feature matrix. Each role guide links into them rather than duplicating the content.
- Operator runbooks [docs/ansible-remote-deploy.md](#) and [docs/break-glass-admin.md](#) are referenced from the Admin and Superuser guides respectively.
- PDF [docs/CLOACK_User_Guide.pdf](#) concatenates all 5 role guides with a TOC; the developer-facing [docs/CLOACK_Documentation.pdf](#) covers the technical architecture instead.



How each role guide is structured

Every role doc follows the same 5 sections so you always know where to look:

1. Who you are + scope 2-3 sentences clarifying which permissions and realms this role typically holds, and which adjacent roles to read if you might be in the wrong place.
2. Quick start the 5-10 most common tasks for this role as a copy-pasteable checklist.
3. Reference matrices the few tables you'll come back to (partitions, ticket statuses, deploy stages, etc.).
4. Troubleshooting symptom -> likely cause -> fix, indexed by error message.
5. Where to go next links to the deep docs (slurm/README.md, CLOACK.md, etc.) and to the in-app help.

How this directory is maintained

- Hand-authored Markdown lives in **docs/roles/*.md**.
- HTML mirrors in **docs/roles/*.html** are regenerated by **images/generate_docs_html.py** (run via the **/docs** skill).
- PDF in **docs/CLOACK_User_Guide.pdf** is regenerated by **python3 .claude/skills/pdf/generate_cloack_pdf.py mode users.**
- The **/cloack** skill picks all of this up automatically and pushes to the public GitHub Pages mirror at **cloack.github.io/jhu-arch** (after the Phase 0 GitHub org transfer lands).



User -- Researcher running jobs

Who you are + scope

You are a researcher with an ARCH Portal account: a JHU faculty / staff / student authenticating with JHED, an external collaborator with an **ext-*** username, or a Schmidt Sciences member with an ssci-*** username**. Your day-to-day is submitting Slurm jobs, checking quotas and bills, and managing the data in your own **/home/** and **/scratch/**.

If you also manage allocations for a project (request a new paid allocation, set Weekly Caps, add other users), read **pi.md** too.

If you are a member of the helpdesk team handling tickets, see **staff.md** instead.

Quick start

1. First-time login

Username pattern	Login URL	How you authenticate
JHED ID (e.g. `rdesouz4`)	the JHU portal card	Entra Federation -> MFA from JHU (no separate password)
`ext-jsmith`	the JHU portal card	LDAP password (set during registration) + TOTP (Keycloak account console)
`ssci-jsmith`	the Schmidt portal card	LDAP password (set during registration) + TOTP (Keycloak account console)

If you do not have an account yet:

- JHED users no registration form; just sign in via JHU SSO and the account is created on first login.

- **ext-*** users** open /user/jhu/register/ on the portal, fill in the form (the username must start with **ext-**), then wait for your PI to add you to their project / allocation.

- **ssci-*** users** same flow but on /user/schmidt/register/, and the username must start with **ssci-**.

After your first successful login a banner at the top of every page prompts you to enrol TOTP in the Keycloak account console. Click the banner link, scan the QR code with Google Authenticator / Authy / Microsoft Authenticator, and you are done. The banner disappears within 30 minutes once the periodic sync picks up the new credential.



2. SSH to a login node

```
ssh <username>@<login-node-host>
```

Authentication on login nodes uses the same identity as the portal:

- JHED [browser-based Device Flow](#) opens automatically; approve in your browser, complete MFA in Entra, the SSH session unblocks.
- **ext-***** / **ssci-***** / **cn=admin** [terminal](#) asks for your LDAP password and then a TOTP code (2 PAM prompts via **pam_kc_ssh.so**).

3. Submit your first job

```
sbatch --partition=med --account=<your-account> --time=00:30:00 \
--wrap='hostname && date'
```

The Slurm CLI Lua filter prompts you to confirm the cost estimate before the job is admitted. Press **y** to confirm or **n** to abort.

For longer jobs, use a script:

```
#!/bin/bash
#SBATCH --partition=high
#SBATCH --account=rdesouz4_proj1
#SBATCH --time=04:00:00
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --job-name=myrun
#SBATCH --output=/home/jhu/rdesouz4/logs/%x-%j.out

module load anaconda
python my_analysis.py
```

4. Check your usage and balance

```
squeue -u $USER          # currently queued or running jobs
sacct -u $USER --starttime=now-7days # last week's accounting
sbalance                 # core-hours used vs the allocation cap
sbalance --week           # weekly $ cap remaining (paid allocations)
```

sbalance reads from ColdFront's allocation attributes [the same](#) numbers the portal's Allocation Detail page shows.

5. Manage data

Path	Quota	Backup	Cost
------	-------	--------	------



`/home/<aff>/<username>`	Small, per-user	Yes (snapshot)	Included
`/home/<aff>/<username>/scratch <pi>`	Symlink into PI's scratch -- same quota as `/scratch`	No	Included
`/scratch/<aff>/<pi>`	Set per allocation (`storage_tb`)	Optional (`backup_tb`)	`\$0.025 / TB / month` storage + `\$0.015 / TB / month` backup
`/scratch/<aff>/<pi>/<project>`	Same quota as parent	Inherits parent	Same

Storage and backup are billed per-month regardless of whether you run jobs. Storage charges show up on your PI's billing report under "Storage" / "Backup" lines.

Reference matrices

Slurm partitions

Partition	Nodes	Max wall	Default?	When to use
`scavenger`	`cn-01`	3 days	**Yes**	Free, preemptible. Default partition. Anything you can re-run if killed.
`low`	`cn-01`	3 days	No	Paid, lowest CPU price. Single-node jobs that tolerate slower scheduling.
`med`	`cn-01`, `cn-02`	3 days	No	Paid, mid-tier CPU price. 2-node jobs.
`high`	`cn-02`, `cn-03`	3 days	No	Paid, highest CPU price. Fast scheduling, larger memory nodes.
`premium`	`cn-01`, `cn-02`	7 days	No	Paid, 7-day wall, all CPU nodes. No `interactive` QoS.
`a100`	`gpu001`	1 day	No	Paid GPU partition (NVIDIA A100).
`i40s`, `h100`, `h200`	varies	varies	No	Paid GPU partitions per hardware generation.

QoS (job class)

QoS	Priority	Max wall	Notes
`scavenger`	low	3 days	Free. Job is preempted if a paid job needs the resource.
`interactive`	1000	4 hours	Higher priority, capped 4 h. Excluded from `premium`.
`normal`	normal	per partition	Default for paid allocations.
`premium`	high	per partition	Higher priority on `premium` partition.
`ssci`	normal	per partition	Free QoS for `ssci-*` accounts on paid partitions (Schmidt Sciences funded).

Cost (per the live rates.lua shipped in the cluster)

Item	Full rate	Overhead-only
CPU service unit (per core-hour)	\$0.35	\$0.10
GPU-hour	\$1.20	\$0.40
GPU partition CPU service unit	\$0.45	\$0.15
Storage	\$0.025 / TB / month	--
Backup	\$0.015 / TB / month	--

GPU-only billing rule when a job uses any GPU, only its GPU charges go into the bill total; the CPU hours of that job are reported for transparency but excluded from the amount due. Pure-CPU jobs



charge CPU as expected.

CLI cheat-sheet

Command	Use
<code>`sinfo`</code>	Show partition / node status.
<code>`squeue`</code>	List queued + running jobs (use <code>`-u \$USER`</code> to filter).
<code>`sbatch <script>`</code>	Submit a batch job.
<code>`srun --pty bash`</code>	Interactive shell on a compute node.
<code>`scancel <jobid>`</code>	Cancel one of your jobs.
<code>`sacct --starttime=...`</code>	Past job accounting (state, exit code, hours).
<code>`sbalance`</code>	ColdFront-side usage vs allocation cap.
<code>`sbalance --week`</code>	Weekly \$ cap remaining for paid allocations.
<code>`seff <jobid>`</code>	Efficiency report after a job finishes.
<code>`stree --full`</code>	Visualise the Slurm account hierarchy you sit under.
<code>`sqos`, `sassoc`, `sfeatures`</code>	ARCH-specific helpers (QoS overview, your associations, partition features).

Troubleshooting

Symptom	Likely cause	Fix
<code>`Permission denied (publickey)`</code> on SSH	Public-key auth disabled by default; you must complete the OTP flow	Re-SSH and follow the terminal prompts (password + TOTP, or browser Device Flow) -- see [docs/break-glass-admin.md](../break-glass-admin.md) for the
Banner says "Activate TOTP" but I already enrolled	Cache up to 30 min -- the <code>`sync_totp_status`</code> task polls Keycloak periodically	Wait 30 min, or hit <code>`/user/user-profile/`</code> to trigger a refresh; if it persists open a helpdesk ticket
enrolled error: AssocMaxCpuMinutesPerJobLimit	Your weekly \$ cap or core-hour cap is exhausted on that allocation	<code>`sbalance --week`</code> to confirm; switch to <code>`scavenger`</code> (free), or ask your PI to raise the Weekly Cap
submit error: Invalid account or account/partition	The account you passed (e.g. <code>`rdesouz4_proj1`</code>) has no QoS that admits jobs on that partition	Use <code>`sassoc`</code> to see which <code>`account x partition`</code> pairs you can submit to
job status in PD with reason	Your account's Weekly Cap on <code>`billing`</code> TRES was reached	<code>`sbalance --week`</code> to confirm; switch partitions / wait for the weekly window to reset / ask the PI to raise the cap
QoS Billing rule job with "accept_cost not set"	A non-CLI submission path (e.g. OnDemand custom app, slurmrestd direct) did not inject <code>`comment=accept_cost`</code>	Re-submit via <code>`sbatch`</code> / <code>`srun`</code> from the login node, or set <code>`--comment=accept_cost`</code> explicitly
Storage charge appears on bill with 0 hours run	Storage and backup are charged monthly regardless of compute activity	This is by design; see the "Storage / Backup billing without jobs" note above
GPU job's CPU hours appear in the report but not in the total	GPU-only billing rule (this page's Cost table)	This is by design -- GPU jobs charge GPU only; CPU hours are shown for transparency
<code>`Permission denied`</code> editing	The directory belongs to the PI, not you	Ask your PI to add you to the project / allocation in ColdFront

``/scratch/jhu/myproject``

Where to go next

- [/user/help/](#) in-app help with live updates.
- [slurm/README.md](#) full Slurm reference

(cost enforcement, hierarchy, weekly cap details, daemons in depth).



- **pi.md** if you also manage a project.
- **README.md** the role index.



PI -- Managing projects, allocations, and billing

Who you are + scope

You are a Principal Investigator with an ARCH Portal account that has the PI flag on your **UserProfile**. You own one or more Projects, request Allocations against them (default scavenger / paid Pay-Per-Use / condo Hardware-Credit), set Weekly Caps, add users to your projects, and read the billing report monthly. Authentication-wise you are a normal user [read user.md](#) for the login flow and the job-submission basics.

If you also operate the cluster (deploy services, configure Slurm) you are a different role [read admin.md](#). If you handle helpdesk tickets you are also Staff [see staff.md](#).

Quick start

1. Accept the Terms of Service

Before you can do anything as a PI you must accept the current ToS. You will see a banner at the top of every portal page until you do. Click the banner -> check the box -> submit. ToS acceptance is also required for every user you add to your projects.

2. Create a project and request an allocation

```
/project/create/          -> create a new project
/project/<id>/            -> project detail page
  Click "Request Resource Allocation" -> choose a resource (Cluster)
```

A project with empty / **default** abbreviation is the scavenger-only default project: it cannot host paid allocations. To request a paid allocation, give the project a real abbreviation (**proj1**, **alpha**, etc.) when you create it.

For paid allocations, fill in:

- IO Number [your funding source](#). Allocations without an IO number are treated as no-charge regardless of partition.



- storage_tb / backup_tb if you need storage and/or backup.
- Weekly Cap (\$) or Weekly Cap (hours) to bound spend.

3. Add users to a project

```
/project/<id>/ -> "Add Users"
```

Users must already have an ARCH Portal account (registered via [/user/jhu/register/](#) or [/user/schmidt/register/](#)) and they must have accepted the current ToS. Once you add them, the Slurm associations provision in real time via the post-save signal [they can submit](#) jobs against the allocation immediately.

4. Set or update a Weekly Cap

```
/allocation/<id>/ -> click the Weekly Cap field -> edit -> save
```

- Dollar cap takes precedence over hours cap if both are set.
- One cap is enforced per allocation (so users sharing the allocation share the cap).
- The gauge below the Core Usage gauge shows current spend vs cap.

5. Read the billing report

```
/billing/report/ -> all-PI view (staff-scoped)
/billing/report-by-pi/ -> per-PI view, your projects only
/user/user-profile/ -> "My Usage & Billing" -> PDF download
```

Storage and backup are billed monthly regardless of compute activity.

GPU jobs charge GPU only [their CPU hours appear for transparency but](#) are excluded from the total due (the GPU-only billing rule, sourced in [billing_views.calculate_allocation_charges](#)).

Reference matrices

Allocation lifecycle

State	Set by	What you can do
New	PI on request	Edit IO / storage / backup; cancel before review
Pending review	Auto on submit	Cannot edit; ColdFront staff approve / deny
Active	Staff approval	Add users, set Weekly Cap, submit jobs
Renewal Requested	PI	Same as Active, with a renewal note for staff



Expired / Archived	Auto on end_date	Read-only; usage history preserved
--------------------	------------------	------------------------------------

Billing model decision table

If...	...pick
Default scavenger PI project (no abbreviation, no IO)	**No Charge** -- auto-set by `ensure_allocation_billing` signal
Schmidt PI (`ssci-*`)	**No Charge** -- funded by the Schmidt programme
Paid project, IO number, no upfront hardware purchase	**Pay-Per-Use**
Paid project, IO number, you bought hardware credit	**Condo** (Hardware-Credit holder; `credits_used` gated by `billing_type=credit_holder`)

Cost Center setup (multi-IO)

You can pool multiple IO numbers under a single project / allocation:

```
/billing/cost-centers/          -> manage your Cost Centers
-> "Add Cost Center"
-> Link Cost Center to a project  -> ProjectCostCenter M2M
-> Assign a Cost Center to an allocation -> AllocationCostCenter M2M
```

ColdFront enforces the Slurm gate: a paid JHU allocation without a pay-role Cost Center linked at the project AND allocation level is silently skipped by **slurm_sync** (no Slurm account created).

Weekly Cap: dollars vs hours

Cap	Units	When to use
Dollar cap	`\$` (converted to `GrpTRESMins=billing=X` via `\$X x 1000 x 60`)	Most allocations. Predictable monthly spend bound.
Hours cap	core-hours	Hardware-Credit / academic allocations where dollars are not the budget unit.

Per partition the dollar value translates differently because of **TRESBillingWeights**. A **med** partition with **CPU=1.0** costs more per core-hour than **low** with **CPU=0.5**.

Troubleshooting

Symptom	Likely cause	Fix
Banner says "Accept Terms of Service" but I already accepted	Old ToS version superseded by a new one	Visit `/user/user-profile/` and accept again; everything you provisioned before stays valid
Adding a user to a project does nothing visible in Slurm	They have not accepted ToS yet -- the post-save signal skips provisioning until	Email the user with the ToS link; once they accept, the immediate-provisioning signal fires
"Request Resource Allocation" button is not shown	the project is a `default` project (no abbreviation)	Create a new project with a real abbreviation; default projects are scavenger-only by design
Allocation is Active but jobs fail with `Invalid account`	The Slurm account hasn't been provisioned yet, or the user's ToS gate has not cleared	Wait for the next `sync_slurm` (every 15 min) or trigger it via Admin actions on the allocation



Storage charge is \$0 but I set <code>`storage_tb=0.20`</code>	<code>`0.20 TB x \$0.025 = \$0.005`</code> , which rounds to \$0.00	This is expected at small sizes; ratio is correct
Billing report shows nothing for the past month	The "Computing Billing" button on <code>`/billing/report/`</code> was never pressed	This is by design: <code>`SlurmAccountUsagePeriod`</code> rows are written only when staff explicitly clicks Compute Billing
Hardware Credit pool reports 0 hours despite usage	The allocation's <code>`billing_type`</code> is not <code>`credit_holder`</code>	Set <code>`billing_type=credit_holder`</code> on the AllocationBilling row via Django Admin, or via the Project Detail page if you have staff

scope

Where to go next

- [/user/help/](#) in-app help with live updates.
- [coldfront/BILLING_SETUP_GUIDE.md](#)

[full billing reference \(rates, periods, invoices, Cost Centers\).](#)

- [user.md](#) login + day-to-day job submission.
- [README.md](#) the role index.



Staff -- Helpdesk operator handling tickets

Who you are + scope

You are a Helpdesk staff member with the **helpdesk-staff** LDAP group on your account (or, for team leads, **helpdesk-admin**). You triage the queue, work tickets through the skill-based assignment matrix, escalate on FQR / SLA breaches, and read the analytics dashboards. You authenticate the same way every other user does see [user.md](#) for the login flow.

If you also approve allocations or run cluster ops, you are also a Superuser read [superuser.md](#). If you author or read tickets without working them, you are a regular **user.md**.

Quick start

1. Triage a new ticket

```
helpdesk -> /helpdesk/                -> dashboard with open tickets
-> click ticket -> ticket detail view
-> Set Priority, Queue, Owner
-> Assign by skill                    -> use the skill-match panel on the right
```

The ticket view banner shows urgency (SLA / FQR / strike / overdue).

The right-hand context sidebar shows the submitter's past tickets, their PI / project / department, GPU / partition references found in the title / description, similar open tickets, and suggested collaborators ranked by skill match.

2. Respond + classify

```
"Send email as <ResponseTemplate>"    -> picks a template
-> {name}/{your_name}/{issue}/{ticket_number}/{survey_url}/{signature}
    are auto-substituted
-> choose the email class (FQR / strike 1-3 / LQR)
-> server-side scrub catches any token you didn't fill in
```

The classification advances the **TicketFollowUpTracker** and feeds the analytics dashboard. Internal notes use **public=False**; customer



emails use **public=True** collaborators are forced to internal-only.

3. Escalate

```
ticket -> "Escalate" action
        -> applies the StaffMember.policy
            (3-strike, FQR / SLA breach)
        -> reassigns to the next staff in
            the routing matrix
        -> logs the routing decision
            for audit
```

/extensions/analytics/routing/ shows the live decision audit trail.

4. Read the analytics

```
/dashboard/
        -> 3 tabs:
            * Department / Project
            * PI
            * Resource (GPU / CPU / Storage / Backup)

/reports/
        -> 8 stock helpdesk reports rendered
            inline as Chart.js bar / line charts

/extensions/skill-matrix/
        -> skills x categories grid
/extensions/calendar/
        -> workload calendar
/extensions/schedule/
        -> engineer x day x slot grid +
            Time Allocations
```

5. Run a public feedback ticket through the Feedback queue

Tickets created via the public submit form with a category whose **show_on_submit=True** are auto-routed to the Feedback & Suggestions queue (slug **feedback**, owner = **ARCH_SUPERUSERS[0]**). They do not pollute General Support review them on a separate cadence.

Reference matrices

Ticket statuses

ID	Status	Open?	Notes
1	Open	Yes	New, untouched.
2	Reopened	Yes	Customer or staff reopened after Resolved / Closed.
3	Resolved	No	Customer can still reply.
4	Closed	No	Locked.
5	Duplicate	No	Customer redirected to canonical ticket.
6	**Archived**	**No**	**ARCH Portal addition.** Resolved / Closed -> Archived. Single exit: Archived -> Reopened.

Priority + SLA windows



Priority	First Quality Response (FQR)	SLA
1 -- Critical	30 minutes	4 hours
2 -- High	2 hours	1 business day
3 -- Normal	1 business day	3 business days
4 -- Low	2 business days	5 business days
5 -- Very Low	best-effort	best-effort

Breaches trigger escalation per **StaffMember.policy** (3-strike rule shipped by default).

Skill matrix routing

Field	What it does
<code>`StaffSkill(technology_domain, ticket_category)`</code>	A staff's competence per (domain, category) pair
<code>`cognitive_process`</code>	Bloom's taxonomy 1-6; **SME threshold** is the singleton <code>`HelpdeskSMEConfig.cognitive_threshold`</code> (default 6 = Evaluation)
<code>`is_mentor`</code>	Admin override per (staff, category); promotes a staff to SME regardless of <code>`cognitive_process`</code>
<code>`TicketCategory.show_on_submit`</code>	Surface this category on the public submit form
<code>`TicketCategory.default_queue`</code>	Auto-route the ticket to a specific queue (e.g. <code>`feedback`</code>)
<code>`TicketCategory.default_assignee`</code>	Auto-route to a product owner

Helpdesk extensions

URL	What it is
<code>`/extensions/skill-matrix/`</code>	Skills x categories grid; admin-editable
<code>`/extensions/calendar/`</code>	Workload calendar (assigned hours per staff per week)
<code>`/extensions/schedule/`</code>	Engineer x day x 30-min-slot grid + Time Allocations
<code>`/extensions/schedule/request/`</code>	Staff submits a weekly allocation request
<code>`/extensions/schedule/my-requests/`</code>	Staff's own request history
<code>`/extensions/schedule/admin/requests/`</code>	Admin reviews + approves / denies requests
<code>`/extensions/deletion-requests/`</code>	Superuser queue for justified ticket deletions
<code>`/extensions/trello/`</code>	Trello browser (Kanban / Cards / Lists / Gantt)
<code>`/extensions/analytics/routing/`</code>	Routing decision audit dashboard

Troubleshooting

Symptom	Likely cause	Fix
New status 6 (Archived) does not show on a tab I expected	Archived is not in <code>`OPEN_STATUSES = (1, 2)`</code>	Archived is a closed status by design -- it shows on the All / Closed tabs only
Skill matrix shows 0 mentors despite recent edits	<code>`hd_sme_threshold`</code> cache TTL = 5 min	Wait, or <code>`cache.delete("hd_sme_threshold")`</code> from the Django shell
Customer email goes out with <code>`{name}`</code> not substituted	The classification dropdown was bypassed (typed directly into the editor)	Pre-save scrub catches this and re-renders the template; review the ticket and resend if needed
Public feedback ticket routed to General Support	The category does NOT have <code>`default_queue`</code> set	Edit the <code>`TicketCategory`</code> in Django Admin and set <code>`default_queue=Feedback`</code>
Collaborator cannot add a comment	Server-side guard: collaborator can only add <code>`public=False`</code> notes	This is by design; only the owner sends customer emails



Reports page is slow	8 sub-reports rendered inline as Chart.js	The two open-by-default cards (Queue-by-Status, User-by-Status) are the hot path; collapse the rest if needed
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Where to go next

- [/staff/help/](#) in-app help (Staff Help Page).
- [/extensions/](#) admin UI for skill matrix, schedule grid,

analytics, deletion-request queue.

- [superuser.md](#) if you also approve allocations

or run cluster ops.

- [README.md](#) the role index.



Superuser -- ColdFront and Django admin

Who you are + scope

You have `is_superuser=True` on your Django **User**. You typically also belong to the LDAP `cn=admin` group (cluster sudoers) and the `helpdesk-admin` group (helpdesk privileges). You approve / deny allocations, drain or resume Slurm nodes, reset stale OTP credentials, rotate the local `/admin/` break-glass password, and read the qcluster audit log for every async task.

You typically also operate the cluster (deploy via `start.sh`, manage Ansible bundles) for that side read `admin.md`. For helpdesk-side work see `staff.md`. You authenticate via Entra Federation if you are JHED; if you are an `ext-***` user you use TOTP via the Keycloak account console (same as every other regular `user.md`).

Quick start

1. Approve / deny an allocation request

```
/admin/allocation/allocation/?status__name=New
-> open the allocation
-> review IO, storage, backup, Weekly Cap
-> action "Approve" or "Deny" (deny requires a reason)
```

Approval triggers `on_allocation_activated`, which provisions the Slurm account hierarchy via REST in real time. ToS gate still applies: PI association is created only if they have accepted the current ToS.

2. Drain / resume a Slurm node

```
/admin/arch_sync/slurmnode/
-> select node -> action "Drain selected nodes" / "Resume"
-> reason is logged on the SlurmNode row
```

Behind the scenes this calls `slurmrestd` (or `scontrol` as fallback) on the right cluster. The action is also surfaced on the live



/cluster// page for staff with the cluster in their scope.

3. Reset a stale OTP credential

```
docker exec keycloak bash -c '/opt/keycloak/bin/kcadm.sh ...'
```

See the recipe in **keycloak/README.md**

"Clear a stale OTP credential" [it requires:](#)

- **kcadm config credentials** once per session (writes **/tmp/kcadm.config**).
- **kcadm get users -q username=\$USER** to fetch the user id.
- **kcadm delete users/\$UID/credentials/** to drop the OTP row.

Alternative [force the user to re-enrol on their next login:](#)

```
kcadm update users/$UID --config /tmp/kcadm.config -r jhu \
-s 'requiredActions=["CONFIGURE_TOTP"]'
```

4. Rotate the break-glass /admin/ password

```
bash scripts/rotate-admin-password.sh
```

Generates a 32-char password, applies it via **manage.py changepassword**

inside the coldfront container, appends an entry to the gitignored audit log, and prints the password once for paste into the team vault.

See **docs/break-glass-admin.md** for the full runbook (vault location, quarterly rotation cadence, **admin_login_alert_task** alert wiring).

5. Read the qcluster audit trail

```
/admin/django_q/task/
/admin/?app=admin&model=logentry -> Home -> Administration -> Log entries
```

Every **async_task()** call uses the **task_completion_hook** so the admin LogEntry table records when each scheduled task ran, success or failure, duration, and result snippet.

Reference matrices



Scheduled tasks (django-q)

Task	Schedule	Purpose
`sync-ldap-daily`	02:00 daily	Full LDAP sync (PIs, members, groups, home dirs).
`sync-slurm`	every 15 min	Safety net for the signal-based real-time provisioning.
`collect-sacct-periodic`	every 15 min	Pull completed jobs from sacct into `SlurmJob`.
`check-core-hours-reset`	01:00 daily	Trigger scheduled period resets on allocations.
`cleanup-task-queue`	hourly :30	Purge stuck / old qcluster tasks.
`sync-totp-status`	every 30 min	Mirror Keycloak TOTP enrolment state into `UserProfile.totp_enrolled`.
`refresh-slurm-jwt`	every 6 h (:17)	Re-issue the slurmrestd JWT to `/etc/slurm/slurm_jwt.token`.
`sync-bare-metal-topology`	every 10 min	`import_slurm_cluster --update` per bare-metal cluster -- Service Health reflects newly-added nodes within 10 min.

LDAP privilege groups (per affiliation tree)

Group	UID/GID	What it grants
`cn=admin,ou=Groups,ou=jhu`	reserved	Cluster sudoers; `sudo` on every Slurm node. Also auto-imports as Django `is_staff=True` via `seed initial_data.sync_admin_group to is_staff`.
`cn=helpdesk-admin,ou=Groups,ou=jhu`	per affiliation	Helpdesk admin (manage queues, change ticket status, edit categories).
`cn=helpdesk-staff,ou=Groups,ou=jhu`	per affiliation	Helpdesk staff (work tickets, classify, reply).
Cluster (LDAP admin) Django Group	n/a	Mirrors `cn=admin` membership for the Django side. Manage at `/admin/auth/group/`.
`ARCH_SUPERUSERS` env var	n/a	Comma-separated usernames auto-promoted to `is_superuser=True` when they land in `cn=admin`.

Cluster Scope + Affiliation Scope (delegated staff admin)

Scope	M2M target	Controls
Cluster Scope	`Resource` (type Cluster)	Drain / resume nodes, view cluster detail, regenerate the per-cluster Ansible bundle
Affiliation Scope	`Affiliation`	Visibility into users, PIs, projects, allocations of those affiliations only (Manage Users page, Promote PI, scoped list views)
`elevate_to_superuser`	bool	Lets a non-superuser staff approve / deny allocations within their scope

Common Django Admin actions

Path	Action	What it does
`/admin/allocation/allocation/`	Approve / Deny	Approves an allocation, fires real-time Slurm provisioning
`/admin/arch_sync/slurmnode/`	Drain / Resume	Slurm node state via REST
`/admin/arch_sync/slurmcluster/`	Action "Action import from slurmrestd" / "Update from slurmrestd" / "Check drift"	Mirror Slurm topology into ColdFront DB
`/admin/user/userprofile/`	Edit	Set `is_pi`, `is_superuser`, `elevate_to_superuser`, `can_view_billing_stats`
`/admin/auth/group/` -> "Cluster (LDAP admin)"	Add member	Publishes to LDAP `cn=admin` via the `m2m_changed` signal



<code>`/admin/django_q/task/`</code>	View	Live scheduled task table
<code>`/admin/?app=admin&model=logentry`</code>	View	Audit trail (all qcluster + admin actions)

Troubleshooting

Symptom	Likely cause	Fix
Allocation approved but Slurm account not created	ToS not accepted by the PI, OR <code>`on_allocation_activated`</code> raised silently	Check the qcluster Log entries; check <code>`TOSAcceptance`</code> rows; manual fallback: <code>`docker exec coldfront python manage.py sync_slurm`</code>
<code>`/admin/`</code> login alert email not received	<code>`ARCH_ADMIN_ALERT_EMAILS`</code> not set in <code>`.env`</code> , OR recipient flagged the SMTP	Set the env var; check <code>`admin_login_alert_task`</code> LogEntry for the dispatch result
<code>`cn=admin`</code> membership change not visible on login	SSSD cache; <code>`sss_cache -E`</code> runs at end of every <code>`run_ldap_sync`</code> but TTL = 60 s	Wait one minute, or run <code>`sss_cache -E`</code> manually on the login node
New user created but no UserProfile	The Postgres trigger <code>`tr_auth_user_after_insert_fn`</code> is missing or out-of-date (e.g. <code>`totp_enrolled`</code> column added after trigger)	Re-run <code>`coldfront/init/ldap_sync_postgres_run_it_once.py`</code> ; see iter-4 fix <code>`1bffe47`</code>
Scheduled task missing from qcluster	django-q scheduler did not pick up the new schedule (entry added but qcluster not restarted)	<code>`docker restart qcluster`</code>
OTP re-enrol flow does not prompt	User does not have <code>`CONFIGURE_TOTP`</code> in their <code>`requiredActions`</code> and <code>`defaultAction=true`</code> only applies to new users	Use the <code>`kcadm update users`</code> snippet from Quick start §3 to add the required action manually
<code>`helpdesk-admin`</code> LDAP membership changed but Django group did not	The signal <code>`_sync_helpdesk_ldap_groups`</code> only fires on <code>`m2m_changed`</code> for <code>`auth.User_groups`</code> -- the inverse direction does not auto-sync	Run <code>`coldfront python manage.py sync_ldap`</code> to reconcile on the next pass
Allocation in "Pending review" forever	Staff scope does not include the requesting affiliation	Add the affiliation to the staff's <code>`StaffResourceAssignment.affiliations`</code> , OR escalate to a superuser

Where to go next

- [/admin/ Django Admin \(Home -> Administration -> Log entries for the qcluster task audit trail\)](#).

- [docs/break-glass-admin.md_full](#)
break-glass [/admin/](#) rotation runbook (vault, audit log, alerts).

- [keycloak/README.md_kcadm_recipes](#)
(OTP credentials, theme re-apply, realm settings inspection).

- [/staff/help/ in-app help with operational notes](#).

- [admin.md_cluster_ops_side](#).

- [README.md_the_role_index](#).



Admin -- Cluster and DevOps operator

Who you are + scope

You operate the ARCH Portal stack on physical hosts: the dev VM on Proxmox, the edumaster bare-metal validation host, and the production portal bare-metal host. You run `./start.sh deploy / start / publish / promote`, you ship images to GHCR, you run **ansible-playbook** for the bare-metal Slurm cluster install, and you own the rotation of the GHCR PATs.

You are the only role that needs read-write access to GHCR. You typically also have **is_superuser=True** in ColdFront for the cluster ops Django side read **superuser.md**. For helpdesk work see **staff.md**. You authenticate the same way every other user does (**user.md**).

Quick start

Five recipes you will run most days. Every command assumes you have already **git clone**'d the repo at the canonical path on whatever host you are on.

1. Local laptop dev (Stage 1)

```
cd ~/arch_jhu_dsai_schmidt
./start.sh deploy dev -y --paid-accounts # full clean + reseed dev stack
bash scripts/verify-deploy.sh           # confirm 8/8 checks green
```

Use this when you want a clean known-good state. It wipes named volumes and reseeds 4 test users (**ext-staff**, **ext-user**, **ssci-staff**, **ssci-user**, password **123**).

2. Build and publish on the Proxmox VM (Stage 2)



```
ssh proxmox-vm
cd ~/jhu-arch
git pull origin dev
./start.sh start dev          # local build, no GHCR
bash scripts/verify-deploy.sh # optional smoke
./start.sh publish            # docker push :dev + :sha-<git-short>
```

Publish refuses with a clear error when the working tree is dirty

(the **:sha-*** tag would lie about content**). **Override with**

tag if you genuinely want a throwaway tag.

3. Promote on edumaster (Stage 3)

```
ssh edumaster
cd ~/jhu-arch
git pull origin dev          # configs / templates only
./start.sh start --from-ghcr dev # docker pull :dev, no build
bash scripts/verify-deploy.sh  # mandatory pre-flight
./start.sh promote           # :dev -> :prod + :rc-<UTC>
# Rollback if a bad promotion slipped through:
./start.sh promote --rollback rc-2026-06-09-1430
```

promote runs **verify-deploy.sh** as a pre-flight gate by default; use

skip-verify only as an emergency override. Never run `./start.sh`

deploy from-ghcr it is hard-refused. Stages 3 and 4 must never

populate test data (the deploy command's job).

4. Deploy on prod portal (Stage 4)

```
ssh portal
cd ~/jhu-arch
git pull origin dev          # configs only
./start.sh start --from-ghcr prod # docker pull :prod, no build
bash scripts/verify-deploy.sh
./start.sh ldap-reinit        # only if LDAP needs re-seed
```

Portal can run Slurm via Docker (pull the slurm images) or via

bare-metal Ansible (see playbook recipes below). The compose overlay

controls which one is active.

5. Bare-metal Slurm via Ansible



```

cd ~/jhu-arch
./start.sh slurm ansible all                # regenerate ansible/ tree
ansible-playbook -i ansible/clusters/<name>/inventory.ini \
  ansible/slurm-server/playbook.yml          # slurmctld + slurmdbd + slurmrestd
ansible-playbook -i ansible/clusters/<name>/inventory.ini \
  ansible/slurm-client/playbook.yml          # slurmd on every compute node
ansible-playbook -i ansible/clusters/<name>/inventory.ini \
  ansible/slurm-verify/playbook.yml          # read-only pre-flight (no changes)

```

slurm-verify is safe to run any time and reports a 14-check matrix

across OS, UIDs, munge, Slurm version, SPANK support, ports, SSSD TTL,
and PAM stack.

Reference matrices

4-stage deploy pipeline

Stage	Host	What you run	Result	Image flow
1 Laptop coding	macOS arm64	<code>./start.sh deploy dev -y --paid-accounts`</code>	Full local dev stack with test data	<code>`arch/<svc>`</code> built locally
1->**2**	Laptop	<code>`git push origin dev`</code>	Source code is on GitHub	--
2 dev VM build	Proxmox Linux amd64	<code>`git pull origin dev`</code>	Code on the builder host	--
2	Proxmox	<code>./start.sh start dev`</code>	Local build matching prod	<code>`arch/<svc>`</code> built locally on amd64
2	Proxmox	<code>./start.sh publish`</code>	<code>`dev`</code> + <code>`:sha-*`</code> pushed to GHCR	<code>`ghcr.io/jhu-arch/cloack/<svc>:dev`</code> + <code>`:sha-<git-short>`</code>
3 edumaster bare-metal	Linux amd64	<code>`git pull origin dev`</code>	Configs / templates on edumaster	--
3	edumaster	<code>./start.sh start --from-ghcr dev`</code>	Stack runs the <code>`:dev`</code> images, no build	<code>`ghcr.io/...:dev`</code> pulled
3	edumaster	<code>`bash`</code>	8-check validation result	--
3	edumaster	<code>./start.sh prod deploy.sh`</code>	<code>`:dev`</code> retagged -> <code>`:prod`</code> + <code>`:rc-*`</code> pushed	<code>`ghcr.io/...:prod`</code> + <code>`:rc-*`</code> pushed
4 portal bare-metal	Linux amd64	<code>`git pull origin dev`</code>	Configs on portal	--
4	portal	<code>./start.sh start --from-ghcr prod`</code>	Stack runs the <code>`:prod`</code> images	<code>`ghcr.io/...:prod`</code> pulled
4	portal	<code>./start.sh ldap-reinit`</code>	LDAP re-seeded if needed	--

(manual)

Direction of artefacts:



```

git origin/dev          GHCR registry
?                       ?
laptop --push--> origin  Stage 2 --push :dev, :sha-*-> GHCR
?                       ?
proxmox --pull--> origin Stage 3 --pull :dev, push :prod, :rc-*-> GHCR
?                       ?
edumaster --pull--> origin Stage 4 --pull :prod
?                       ?
portal --pull--> origin  (portal never pushes)

```

GHCR registry namespace and PATs

Stage	PAT scope	Operations
1 (laptop)	none	git push only
2 (Proxmox)	`GHCR_BUILDER_PAT` (write)	`docker push :dev` + `:sha-*`
3 (edumaster)	`GHCR_BUILDER_PAT` (write)	`docker pull :dev`, `docker push :prod` + `:rc-*`
4 (portal)	`GHCR_CONSUMER_PAT` (read-only)	`docker pull :prod` only

Registry path: **ghcr.io/jhu-arch/cloack/**.

Cross-org topology: the repo lives in the **cloack** GitHub org, packages live in the **jhu-arch** org. The two PATs are fine-grained, scoped to the **jhu-arch** org, and SSO-authorized for Johns Hopkins University Enterprise. Rotation cadence: <= 1 year (fine-grained PAT maximum lifetime).

Image inventory (13 -- 6 cloack apps + 7 slurm; cn-rockylinux is build-only)

Track	Image	Built from	Promoted past `:dev`?
Cloack	`coldfront`, `helpdesk`, `keycloak`, `keycloak-config`, `openldap`, `phpldapadmin`, `postgres`	each service's `Dockerfile`	Yes -- Stage 3 + 4
Slurm (base)	`cn-rockylinux`	`slurm/base/Dockerfile`	**No** -- build dep only, never pulled by runtime hosts
Slurm (services)	`slurmctld`, `slurmd`, `slurmrest`, `slurm-munge`, `mariadb`, `slurmd`	`slurm/<svc>/Dockerfile`	Yes (in Docker-Slurm Mode A) -- Stage 3 + 4

Note that **cn-rockylinux** is consumed at build time by the other Slurm images (each does a **FROM arch/cn-rockylinux**). Once a Slurm service image is pushed, the cn-rockylinux layers travel inside it runtime hosts (Stage 3 + 4) never pull cn-rockylinux as a standalone image.

Stage 4 dual-mode Slurm

Mode	Slurm runs as	What portal pulls	When to use
A -- Docker Slurm	containers	6 cloack apps + 6 Slurm services	All-Docker portal, smaller HPC scale



B -- Bare-metal Slurm	services on real hosts via Ansible roles	6 cloack apps only	Real HPC scale, dedicated compute nodes, kernel-level features
------------------------------	---	--------------------	---

In Mode B the operator runs **ansible/slurm-server/playbook.yml** and **ansible/slurm-client/playbook.yml** against the cluster inventory.

cluster_manager.write_compose_overlay honours

CLOACK_FROM_GHCR=1 to rewrite Slurm image refs to **ghcr.io/...:prod**

in Mode A.

start.sh subcommands by stage

Subcommand	Stage(s)	What it does							
`init [dev]`	1, 2	Bootstrap user/group/dirs/.env. Run once before							
`start [dev]`	1, 2	Builds (art`skip build) and start managed services.							
`start --from-ghcr [dev]`	prod]	3, 4	Pull from GHCR, skip build. Refuses if `docker-compo						
`deploy [dev] -y [--paid-accounts]`	1, 2	Destructive: clean + init + start + populate test data. Hard-refused with	se-ghcr.yml` is missing.						
`publish [--tag <tag>] [--check]`	2	Retag-ghcr`. local `arch/<svc>` as `ghcr.io/...:<tag>` + `:sha-<short>` and push. Refuses dirty							
`promote [--from <tag>] [--rc <label>] [--check] [--rollback]`	3	Rolling tree, run `verify-deploy.sh`, retag + push `:prod` + `:rc-<UTC>`.							
`deploy-init [dev]`	3, 4	Re-initialise OpenLDAP from migration LDIF, force sync with							
`slurm {create\	start\	Stop Front.	restart\	status\	test-jobs \	ansible\	prod-sim}`	varies	Per-cluster Slurm operations. `ansible all` regenerates
`helpdesk {start\	populate\	restart\	stop}`	1, 2	Helpdesk containe				

r
lifecycle.



`doctor`	any	Read-only validation of `.env` <-> realm config <-> TLS cert <-> Host
----------	-----	--

header drift.

Ansible playbooks

Playbook	What it does	When to run
`ansible/slurm-server/playbook.yml`	Compiles Slurm from source, installs slurmctld / slurmd / slurmrestd on the controller groups	Bare-metal install or major Slurm version bump
`ansible/slurm-client/playbook.yml`	Same build stack plus slurmd on every compute node and login node	New compute node, or after `slurm-server`
`ansible/slurm-verify/playbook.yml`	Read-only 14-check pre-flight (OS, UIDs, munge, Slurm version, SPANK, ports, SSSD TTL, PAM stack). No changes applied.	Any time you want a confidence check
`ansible/sssflush/playbook.yml`	Runs `sss_cache -E` on every login node	`sync_ldap` already triggers this per cluster; manual run useful after a large LDIF import
`cloak-deploy` RPM	Thin packaging of the Ansible tree + a `cloak-deploy` CLI wrapper	Sites that prefer an RPM-driven flow over `ansible-playbook` directly

Environment variables you will set on Stage 3 / Stage 4 hosts

Phase 2 (commit **a525237**) added multi-FQDN reverse-proxy support. The

following env vars are opt-in; leave them empty in single-FQDN dev

to keep current behaviour.

Variable	Purpose
`ARCH_PUBLIC_HOST`	The portal FQDN (`portal.arch.jhu.edu`). Used everywhere.
`KEYCLOAK_HOSTNAME_JHU`	JHU realm FQDN (`auth.arch.jhu.edu`). Empty -> single-FQDN.
`KEYCLOAK_HOSTNAME_SCHMIDT`	Schmidt realm FQDN. Empty -> single-FQDN.
`KC_HOSTNAME`	Empty in multi-FQDN; Keycloak uses inbound Host instead.
`KC_HOSTNAME_BACKCHANNEL_DYNAMIC`	`true` in multi-FQDN deploys (trust `X-Forwarded-Host`).
`HELPDESK_SCRIPT_NAME`	`/helpdesk` in prod to route `portal.arch.jhu.edu/helpdesk` to the helpdesk container. Empty -> no prefix.
`CLOACK_IMAGE_TAG`	Auto-set by `--from-ghcr`; override only for rollback (`sha-abc123` or `rc-YYYY-MM-DD-HHMM`).
`CLOACK_FROM_GHCR`	Auto-set by `--from-ghcr`; do not set by hand.

Troubleshooting

Symptom	Likely cause	Fix
`keycloak-config` Error... exit 5` on first deploy	`jq` could not index into an error response on `/auth/flows` (realm not imported yet)	Already fixed in commit `cd35aad`; if reappears, check the `_GHCR_IMAGE_REWRITE_REGEX` regex hasn't been broken by a new image name pattern
`keycloak-config` Error... exit 3` on first deploy	`jq` tried `ascii_lowercase` on a null `displayName` (Keycloak default executions list)	Fixed in `8443758` -- re-pull the latest dev if you see this



`curl: URL malformed` in configure-keycloak.sh	Copied flow has a literal space in its sub-flow alias (`passwordless-browser forms`)	Fixed in `d0dfafe` -- `\${forms_sub_alias// /%20}` URL-encodes the space	
`IntegrityError: null value in column "totp_enrolled"``	Postgres trigger `tr_auth_user_after_insert_fn` was created before the `totp_enrolled`	Fixed in `1bffe47` -- drop the trigger and re-run `ldap_sync_postgres_run_it_once.py`	
Themes do not apply on first deploy (default Keycloak look)	configure_realm_themes ran before `import_keycloak_providers` so PUT returned 404 silently	Fixed in `b7efa22` -- re-runs the idempotent block after realms are imported	
`publish: working tree is dirty`	You have uncommitted source changes	Commit them, or use `./start.sh publish --tag wip-<something>` to bypass the SHA-tag promise	
`deploy --from-ghcr is not allowed`	Stages 3 and 4 must never populate test data	Use `./start.sh start --from-ghcr <tag>` plus `./start.sh ldap-reinit` instead	
`publish: no GHCR auth in ~/docker/config.json`	Builder PAT not loaded	`echo \$GHCR_BUILDER_PAT \	docker login ghcr.io -u <user> --password-stdin`
`--from-ghcr: docker-compose-ghcr. yaml not found`	Phase 0 file missing in the repo on this host	Pull latest dev (`52b709f` introduced the file)	
`pgrep -f "start.sh deploy"`` loop never	The waiter's own command line contains the pattern, deadlock	Track `\$DEPLOY_PID` directly with `wait \$DEPLOY_PID`; see memory `feedback pgrep self match deadlock`	
Slurm overlay still emits `arch/<svc>` after `--from-ghcr`	`CLOACK_FROM_GHCR` was not exported when `write_compose_overlay` ran	Re-run `./start.sh slurm create <name>` after exporting the var	
`configure_realm_from tend_urls` did nothing visible	`KEYCLOAK_HOSTNAME_JHU` / `_SCHMIDT` are empty	Set them in `.env` for multi-FQDN, or leave them empty for single-FQDN -- the function is a no-op then by design	

Where to go next

- [docs/ansible-remote-deploy.md](#) full

bare-metal Ansible deploy guide, RPM install, idempotent re-runs.

- [docs/break-glass-admin.md](#) /admin/

break-glass rotation (vault, audit log, alerts).

- [slurm/README.md](#) full Slurm config

reference (partitions, QOS, weekly cap, CLI tools).

- [slurm/ANSIBLE.md](#) bare-metal Slurm install

via the **cloak-deploy** RPM path.

- [.claude/plans/deploy-pipeline-4-stages.md](#) the design plan behind

the matrix on this page; read when the model needs to evolve.

- [superuser.md](#) ColdFront / Django admin side

(drain nodes, approve allocations, qcluster audit, OTP resets).

- [README.md](#) the role index.